Ollscoil Teicneolaíochta an Oirdheiscirt

South East Technological University

ECADS
Emergency Services Contact & Dispatch System for offline management

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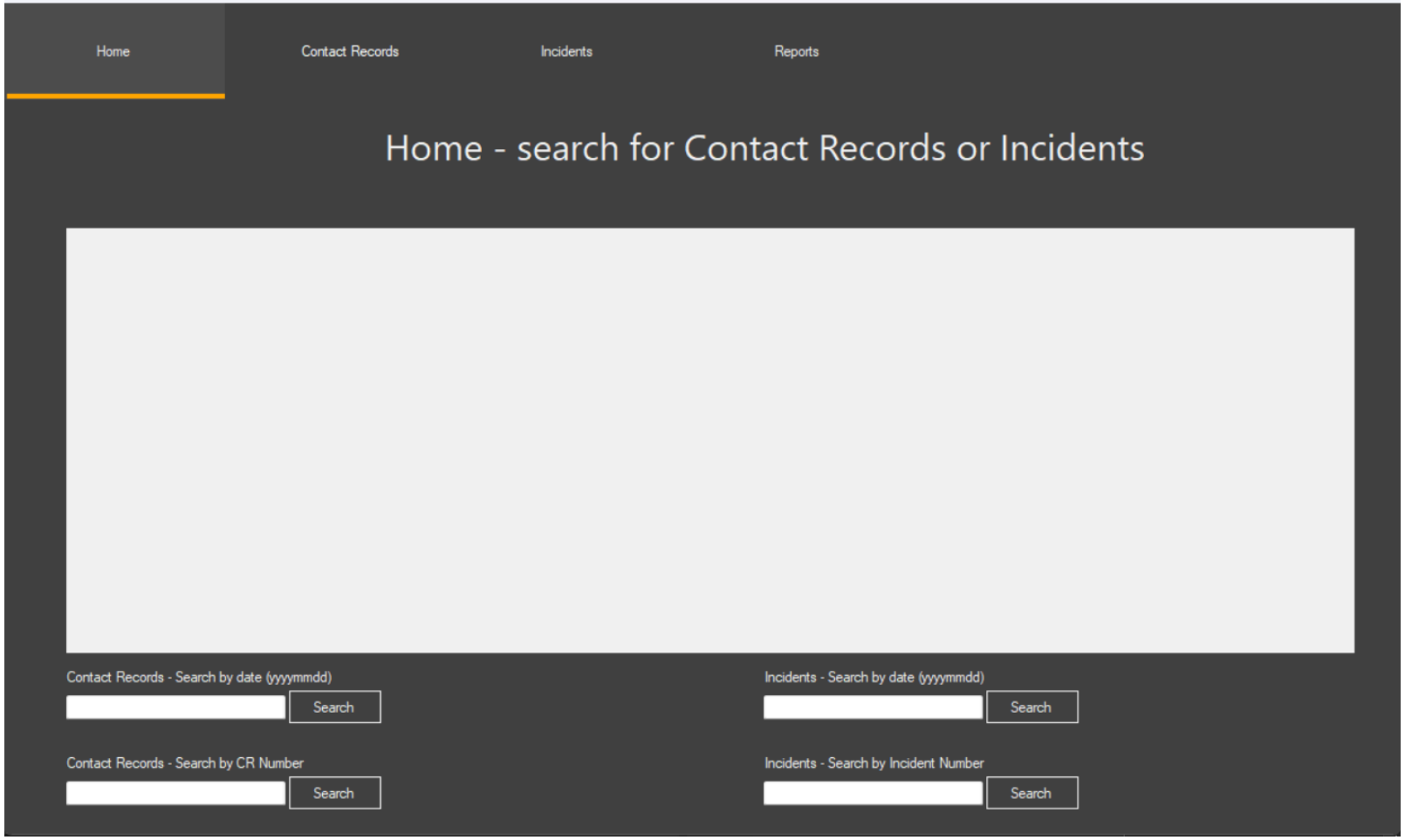
Introduction

- Feargal Mc Loughlin
- 39 Years of age
- Technical Lead on a COTS dispatch system provided by SAAB in An Garda Síochána
- Aim of the project:
  - To capture data in an organised fashion in a “system down” scenario in Control Centres in An Garda Siochána
  - Implement same governance rules in this offline system as are in existence in the main current system, i.e. no ability to open closed issues, no ability to change certain properties that have been entered but the ability to update Incidents with comments and developments.

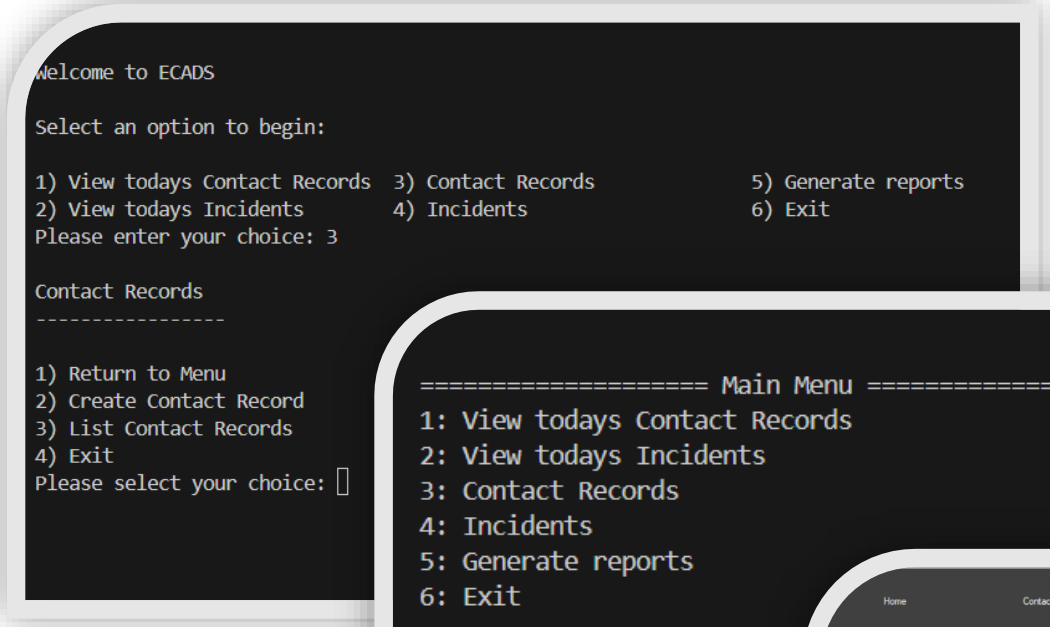
ECADS - Demonstration

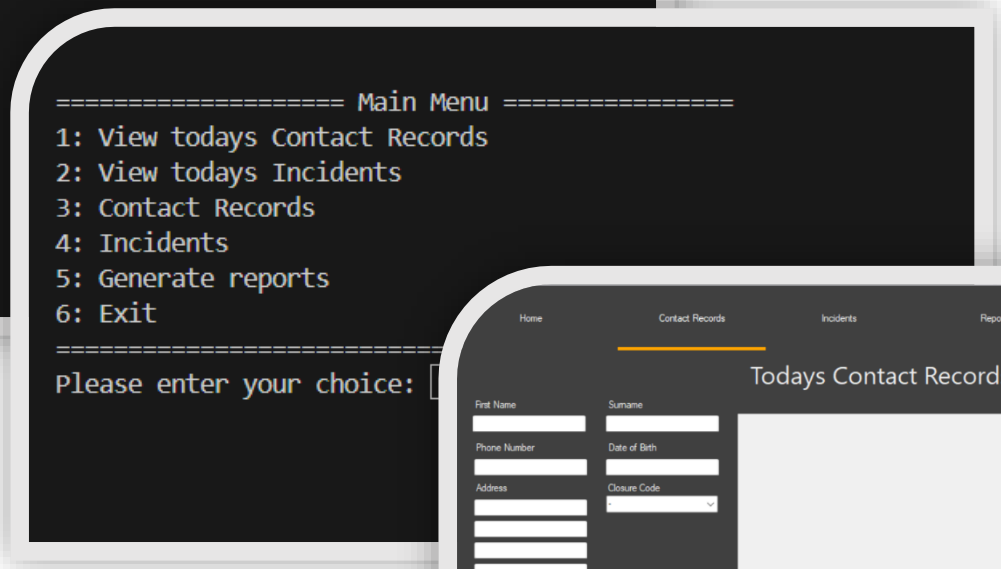


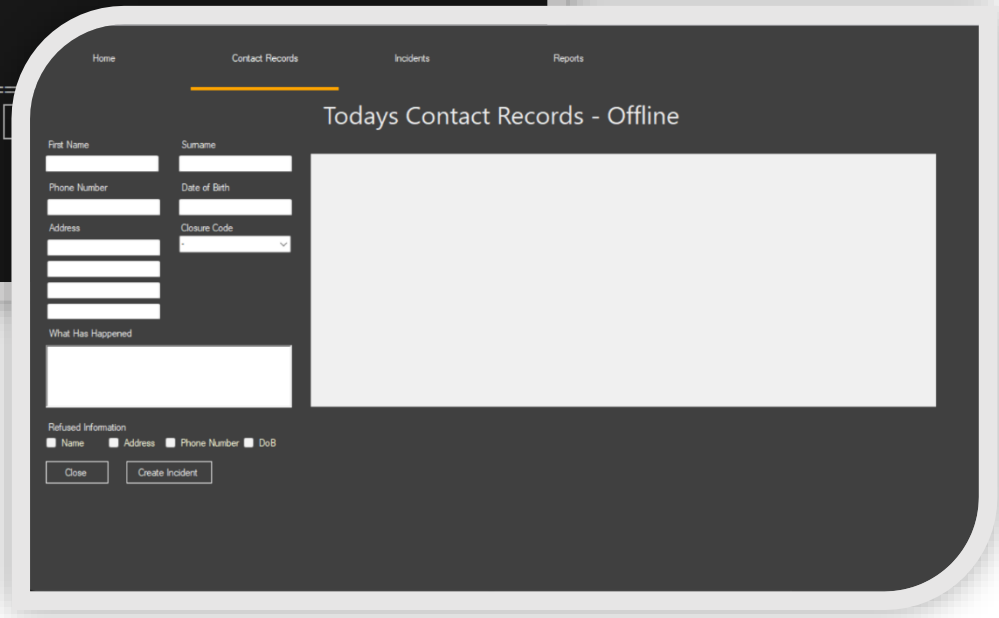
Development

An agile approach was taken towards development, starting with a proof of concept with BASH scripts, progressing to a single PowerShell script and resulting in a Windows Forms fully functional application.

Each iteration went through a feedback process with real users feeding into the features of the application – this allowed for a real “customer-like” development environment to be experienced.







Key Points

From BASH to Window Forms

- Script based CLI style program proved the concept and functionality
- Highlighted the differences in the ease of programming once designed and mapped correctly
- Complexity from a coding side increased as the versions progressed
- Although easy to interact with, the CLI versions were very linear and a clear response in UAT is that a GUI format was required
- C# and Windows Forms allowed for quickly designing a GUI application but more thought was needed to achieve same functionality

Problems to overcome

- Storage of data
  - Treat .txt files as human readable string arrays
- Distinguishing text to ensure areas of importance are highlighted to end user
- End User experience
  - Familiarity
  - Ease of access
  - Ease of Use
  - Data validation
- Data retrieval
- Date merging
- Application maintenance and development

Artificial Intelligence (AI)

Use of AI

- The use of AI was encouraged throughout the information sessions on the project. I myself chose not to engage with AI on this project as I wished to use this experience to learn new languages and problem solve.
- I believe were I to use AI for tis project that I would not have had the insight into the many different ways I used to overcome simple problems that varied between the three different languages used.
- A recent study (Kosmyna et al, 2025) hosted on MIT Media Lab, indicates that over reliance on AI can lead to impaired ability to critically think, analyse and problem solve – all areas I wished to improve upon while undertaking this project. It must be noted however that this study involved engaging with ChatGPT for essay writing.

How AI could be used in future development

- I do see scope for the use of AI in further developing out this project, when different features are called for or when building a more streamlined way in which to incorporate the data into existing databases

Conclusion

- ECADS is a personal project that has real world implications
- Abiding by strict requirements as set down by the customer doesn't curtail functionality, per se – It allows the developer come up with new and thought provoking ways to overcome the obstacles that may in fact hinder what was once thought functionality
- Taking an application through different stages of development means that issues can be identified early on and results in less time being wasted when it comes to more complex programming
- Simplicity is key – get the application to do what you want (functionality) and then refine code afterwards (efficiency & readability)
- Understanding of methods and their use will assist in abiding by DRY principles and help in code maintenance (location of directories and file creation)
- There are always multiple ways to achieve what you want to do – aligning what can be done with the expectations of users / customers and time, will decide in some respect what you will do and how you do it
- An application <u>never</u> has a final release – it only has the most up-to-date version

References

Kosmyna N., Hauptmann E., Tong Yuan Y., Situ J., Liao X., Beresnitzky A.V., Braunstein I. and Maes P. (2025) Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task. *arXiv* Available from: https://doi.org/10.48550/arXiv.2506.08872 [accessed 12 February 2026].